Atomic Lab: A Technical Whitepaper on Agentic RAG and Atomic Knowledge Workflows

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Abstract

Welcome to Atomic Lab. This document serves as both a technical whitepaper and an interactive demo file for our system. The exponential growth of scientific literature presents a dual crisis: the fragmentation of reading inspirations and the difficulty of tracing citations during writing. Atomic Lab solves this by deconstructing literature into "Atomic Knowledge" and leveraging an Agentic RAG (Retrieval-Augmented Generation) architecture. This manual outlines our core technical implementations.

1 Introduction to Atomic Knowledge

Traditional document-centric reading models trap knowledge within isolated PDFs. We advocate for "Knowledge Atomization." Every core argument, experimental result, or methodology is extracted as a semantically self-consistent "Atomic Card."

By passing extracted text through our *Crusher Agent*, non-structured text is automatically categorized into seven domains (Method, Formula, Image, Definition, Viewpoint, Data, Other) and tagged with AI-generated keywords.

2 Core Technologies

2.1 Multi-modal Deep Parsing

Academic literature parsing is notoriously difficult due to dual-column layouts and complex mathematical embeddings. Atomic Lab integrates the advanced **MinerU** parsing engine.

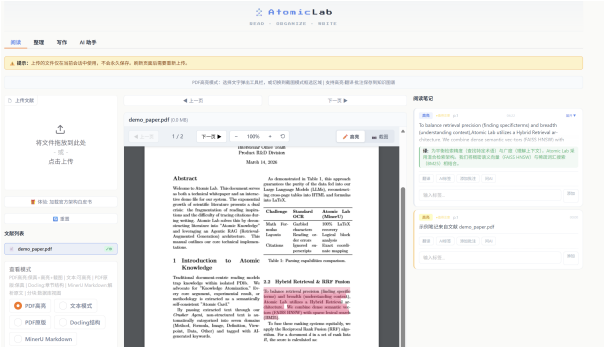

Figure 1: Atomic Lab

As demonstrated in Table 1, this approach guarantees the purity of the data fed into our Large Language Models (LLMs), reconstructing cross-page tables into HTML and formulas into LaTeX.

Challenge	Standard OCR	Atomic Lab (MinerU)
Math Formulas	Garbled characters	100% LaTeX recovery
Layouts	Reading order errors	Logical block analysis
Citations	Ignored superscripts	Exact coordinate mapping

Table 1: Parsing capabilities comparison.

2.2 Hybrid Retrieval & RRF Fusion

To balance retrieval precision (finding specific terms) and breadth (understanding context), Atomic Lab utilizes a Hybrid Retrieval architecture. We combine dense semantic vectors (FAISS HNSW) with sparse lexical search (BM25).

To fuse these ranking systems equitably, we apply the Reciprocal Rank Fusion (RRF) algorithm. For a document d in a set of rank lists R, the score is calculated as:

$$RRF(d) = \sum_{r \in R} \frac{1}{k + r(d)} \tag{1}$$

where $k = 60$ acts as a smoothing constant. This ensures that only knowledge atoms ranking high across multiple strategies are presented to the user.

2.3 GraphRAG & 1-Hop Expansion

Knowledge in Atomic Lab is not flat. Atoms are stored in a dual-track architecture. When a user queries a concept, the system retrieves direct matches and performs a "1-hop expansion" along the knowledge graph (e.g., following shared tags or citation links) to uncover hidden academic causality.

3 The Agentic RAG Workflow

The chat interface of Atomic Lab is not a simple Q&A bot; it simulates a multi-agent "virtual research group" meeting:

- **Seeker:** Executes multi-source retrieval, pulling evidence from document vectors, note cards, and external fallback APIs (e.g., arXiv).
- **Reviewer:** Critically evaluates the retrieved context. If the evidence is insufficient, it rewrites the query and triggers a secondary search.
- **Synthesizer:** Generates the final, logically coherent response, embedding precise, verifiable citations.

4 Conclusion

By integrating deep document parsing, GraphRAG, and Agentic workflows, Atomic Lab provides a seamless *Read-Organize-Write* loop. We invite you to explore the system by highlighting texts, asking questions, and generating your own verifiable academic insights.